

TASK 9 – Record Experimental Performance

1. Objective

The objective of this task is to record and compare the experimental performance of Linear Search, Recursive Binary Search, and Hash Table Search for different input sizes. The performance is evaluated using the number of student records, number of key comparisons, execution time, and preprocessing requirements.

2. Experimental Performance Table

The following table is used to record the measured execution time of each algorithm.

Input Size (n)	Linear Search (seconds)	Binary Search (seconds)	Hash Table Search (seconds)
1,000	_____	_____	_____
10,000	_____	_____	_____
100,000	_____	_____	_____
1,000,000	_____	_____	_____

The execution times should be collected by running all three algorithms using the same dataset and consistent testing conditions. For Hash Table Search, the table construction time should be recorded separately because it is a preprocessing operation.

3. Theoretical Comparison of Key Comparisons

For a successful search, the number of comparisons depends on the position of the target registration number. The following table shows the approximate comparison counts for representative cases.

Input Size	Linear Search (worst case)	Binary Search (approximate worst case)	Hash Search (average expected)
1,000	1,000	10	Nearly constant
10,000	10,000	14	Nearly constant
100,000	100,000	17	Nearly constant
1,000,000	1,000,000	20	Nearly constant

Note: These are theoretical estimates, not measured experimental results. Hash table comparisons depend on the hash function, load factor, and collisions. Binary Search comparison counts are approximate and depend on the implementation.

